Linear Algebra

Lecture 06

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Fundamental Theorem of Linear Algebra

Theorem 1.5.3

If A is $n \times n$ matrix then the following statements are equivalent, that is, all true, or all false

(a) A is invertible.

(b) $Ax = 0$ has only the trivial solution.

(c) The reduced row-echelon form of A is I_n.

(d) A is expressible as a product of elementary matrices.

Proof: We shall prove the equivalence by establishing the chain of implications: $(a) \implies (b) \implies (c) \implies (d) \implies (a)$.

$(a) \implies (b)$**:** Assume A is invertible and let x_0 is the solution of $Ax = 0$, which implies that $Ax_0 = 0$. Multiply both sides of this equation by the matrix A^{-1} gives

$$\begin{aligned} A^{-1}(Ax_0) &= A^{-1}0 \\ (A^{-1}A)x_0 &= 0 \\ Ix_0 &= 0 \\ x_0 &= 0 \end{aligned}$$

This means, $Ax = 0$ has only the trivial solution.

$(b) \implies (c)$**:** We are given that $Ax = 0$ has only the trivial solution and we have to show that the matrix A can be re-written in row-reduced echelon form or I_n. This we can achieve with the idea of elementary row operations which will transform the augmented matrix

$$\begin{bmatrix} a_{11} & a_{12} & \cdots a_{1n} & 0 \\ a_{21} & a_{22} & \cdots a_{2n} & 0 \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots a_{nn} & 0 \end{bmatrix}$$

into

$$\begin{bmatrix} 1 & 0 & \cdots 0 & 0 \\ 0 & 1 & \cdots 0 & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots 1 & 0 \end{bmatrix}$$

If we dis-regard the last column, we can conclude that the matrix is in row-reduced echelon form.

$(c) \implies (d)$**:** We know that the row-reduced echelon form of A is I_n and we want to show that A can be expressed as the product of elementary matrices.
As we know that

> Theorem 1.5.1
>
> If the elementary matrix E results from performing a certain row operations on I_m and if A is an $m \times n$ matrix, then the product EA is the matrix that results when the same row operation is performed on A.

Because we know that with the application of finite number of elementary row operations we can transform a matrix into identity matrix. Each of these E.R.O can be accompanied by multiplying on the left by an appropriate elementary matrix. Thus we can find elementary matrices $E_1, E_2, \cdots E_k$ such that

$$E_k E_{k-1} \cdots E_2 E_1 A = I \tag{1}$$

By using

Theorem 1.5.2

Every elementary matrix is invertible, and the inverse is also an elementary matrix.

This implies that $E_1, E_2, \cdots, E_k$ are invertible. Multiplying both sides of equation (1) on the left successively by $E_k^{-1}, \cdots, E_2^{-1}, E_1^{-1}$ we obtain

$$\begin{aligned} A &= E_1^{-1}E_2^{-1}\cdots E_{k-1}^{-1}E_k^{-1}I_n \\ &= E_1^{-1}E_2^{-1}\cdots E_{k-1}^{-1}E_k^{-1} \end{aligned}$$

This shows that A can be expressed as a product of elementary matrices.

$(d) \implies (a)$**:** We consider that A is written as a product of elementary matrices and want to show that A is invertible. By using

Theorem 1.4.6

If A and B are invertible matrices of the same size, then AB is invertible and

$$(AB)^{-1} = B^{-1}A^{-1}$$

with Theorem 1.5.2, we can see that A is invertible.

Theorem 1.6.1

Every system of linear equations has no solution, or has exactly one solution, or has infinitely many solutions.

Proof: If $Ax = b$ is a system of linear equations, exactly one of the following is true: (a) the system has a unique solution, (b) the system has no solution, or (c) the system has more than one solution. The proof will be complete if we can show that the system has infinitely many solution in case (c).

Assume that $Ax = b$ has more than one solution, and let $x_0 = x_1 - x_2$, where x_1, x_2 are two distinct solutions. Because x_1, x_2 are distinct, the matrix x_0 is nonzero and

$$\begin{aligned} Ax_0 &= A(x_1 - x_2) \\ &= Ax_1 - Ax_2 \\ &= b - b \\ Ax_0 &= 0 \end{aligned}$$

If we know let k be any scalar, then

$$\begin{aligned} A(x_1 + kx_0) &= Ax_1 + A(kx_0) \\ &= Ax_1 + kAx_0 \\ &= b + k0 \\ &= b \end{aligned}$$

This means that $x_1 + kx_0$ is a solution of $Ax = b$. Since x_0 is nonzero and there are infinitely many choices of k, the system $Ax = b$ has infinitely many solutions.

Theorem 1.6.3

Let A be a square matrix.

(a) If B is a square matrix satisfying $BA = I$, then $B = A^{-1}$.

(b) If B is a square matrix satisfying $AB = I$, then $B = A^{-1}$.

We shall prove part (a) and leave part (b) as an exercise. Assume that $BA = I$. If we can show that A is invertible, the proof can be completed by multiplying $BA = I$ on both sides by A^{-1} to obtain

$$\begin{aligned} BAA^{-1} &= IA^{-1} \\ BI &= IA^{-1} \\ B &= A^{-1} \end{aligned}$$

To show that A is invertible, it suffices to show that $Ax = 0$ has only the trivial solution [Fundamental Theorem of Algebra $(a) \implies (b)$]. Let x_0 be any solution of this system. If we multiply both sides of $Ax_0 = 0$ by B, we get

$$\begin{aligned} BAx_0 &= B0 \\ Ix_0 &= 0 \\ x_0 &= 0 \end{aligned}$$

Thus the system of equations $Ax_0 = 0$ has only trivial solution which is only possible if A is invertible.

Re-visiting the Fundamental Theorem of Algebra

Theorem 1.6.4

If A is $n \times n$ matrix then the following statements are equivalent, that is, all true, or all false

(a) A is invertible.

(b) $Ax = 0$ has only the trivial solution.

(c) The reduced row-echelon form of A is I_n.

(d) A is expressible as a product of elementary matrices.

(e) $Ax = b$ is consistent for every $n \times 1$ matrix b.

(f) $Ax = b$ has exactly one solution for every $n \times 1$ matrix b.

Since we proved in Theorem 1.5.3 that (a), (b), (c), and (d) are equivalent, it will be sufficient to prove that $(a) \implies (f) \implies (e)$.

$(\mathbf{a}) \implies (\mathbf{f})$:

Because A is invertible. This means

$$\begin{aligned} A^{-1}Ax &= A^{-1}b \\ x &= A^{-1}b \end{aligned}$$

As the inverse of a matrix is unique, this implies that x is a unique solution for $Ax = b$.

$(\mathbf{f}) \implies (\mathbf{e})$: This is self evident. If $Ax = b$ has exactly one solution for every $n \times 1$ matrix b, then $Ax = b$ is consistent for every $n \times 1$ matrix b.

$(\mathbf{e}) \implies (\mathbf{a})$: If the system $Ax = b$ is consistent for every $n \times 1$

matrix b, then in particular, the systems

$$Ax = \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, Ax = \begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \cdots, Ax = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}$$

are consistent. Let $x_1, x_2, \cdots, x_n$ be the solutions of the respective systems, and let us form an $n \times n$ matrix C having these solutions as columns. Thus C has the form

$$C = [x_1|x_2|\cdots|x_n]$$

As discussed earlier, the successive columns of the product AC will be

$$Ax_1, Ax_2, \cdots Ax_n$$

Thus,

$$\begin{aligned} AC &= [Ax_1|Ax_2|\cdots|Ax_n] \\ &= \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix} \end{aligned}$$

By using the Theorem 1.6.3, it follows that $C = A^{-1}$. Thus A is invertible.